



Computing Functions of Rank-structured or Telescopic Matrices

MS 36 contribution, ILAS, Virginia Tech

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Goal and motivation

1 Introduction

- We want to compute $f(A)$ where A is not necessarily sparse but has low-rank off-diagonal blocks (Banded, Cauchy-like, Hierarchical, ...);
- We want to do it $\mathcal{O}(n)$ and return the output in a structured representation¹;
- Applications: exponential integrators, ODE with non-local operators, functions of Toeplitz matrices², ...

¹Cortinovis, Kressner, and Massei, “Divide-and-conquer methods for functions of matrices with banded or hierarchical low-rank structure”.

²Beckermann, Kressner, and Wilber, *Compression Properties for large Toeplitz-like matrices*.



Hermitian A^5

1 Introduction

- **Hermitian Hierarchically Semiseparable matrices** can be decomposed into a sequence of low-rank updates³;
- This **telescopic decomposition** can be used to compute $f(A)$ efficiently with Krylov spaces.

What's new? We generalize this method to the non-Hermitian case.

Why it matters? $f(A) \approx$ HSS and **displacement**⁴ links classical structured matrices to HSS.

³Levitt and Martinsson, “Linear-Complexity Black-Box Randomized Compression of Rank-Struct. Matrices”.

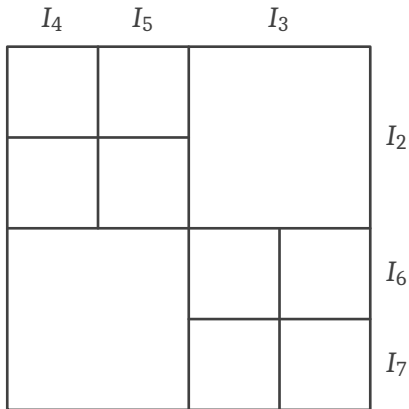
⁴Beckermann and Townsend, “On the Singular Values of Matrices with Displacement Structure”.

⁵Casulli, Kressner, and Robol, “Computing Functions of Symmetric Hierarchically Semiseparable Matrices”.

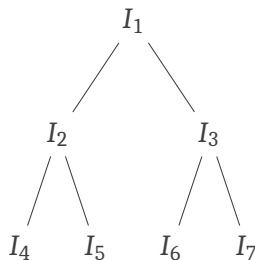


HSS are $\mathcal{H}^2$ with a special block-partitioning

2 HSS matrices and telescopic decomposition



Hierarchical partitioning (*cluster tree*) $\mathcal{T}$ of depth $L = 2$.



$$I_1 = [1, 2, \dots, n]$$

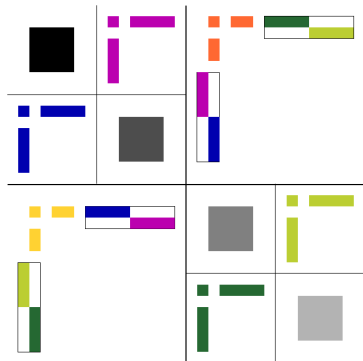
$$I_2 = [1, \dots, \lceil n/2 \rceil], I_3$$

$$I_4 = [1, \dots, \lceil n/4 \rceil], \dots, I_7$$



Hierarchically semi-separable (HSS) matrices

2 HSS matrices and telescopic decomposition



1. Low-rank off-diagonal blocks:

For $\tau, \tau' \in \mathcal{T}$ siblings,

$$A_{\tau, \tau'} = U_{\tau}^{\text{big}} \tilde{A}_{\tau, \tau'} (V_{\tau'}^{\text{big}})^*,$$

where $U_{\tau}^{\text{big}}, V_{\tau'}^{\text{big}}$ have ONC and $\tilde{A}_{\tau, \tau'} \in \mathbb{C}^{r \times r}$.

2. Nested basis structure:

For non-leaf $\tau \in \mathcal{T}$ with children α and β ,

$$U_{\tau}^{\text{big}} = \begin{bmatrix} U_{\alpha}^{\text{big}} \\ U_{\beta}^{\text{big}} \end{bmatrix} U_{\tau}, \quad V_{\tau}^{\text{big}} = \begin{bmatrix} V_{\alpha}^{\text{big}} \\ V_{\beta}^{\text{big}} \end{bmatrix} V_{\tau},$$

where U_{τ}, V_{τ} have ONC.



Standard telescopic decomposition

2 HSS matrices and telescopic decomposition

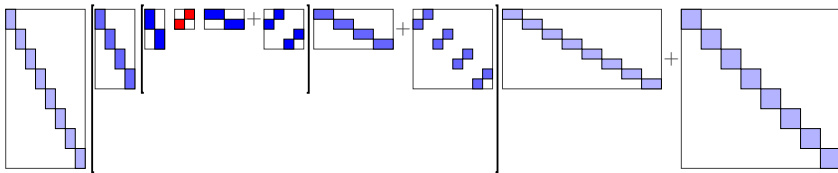
$$A = \underbrace{\begin{bmatrix} A_{44} & & & \\ & A_{55} & & \\ & & A_{66} & \\ & & & A_{77} \end{bmatrix}}_D + \underbrace{\begin{bmatrix} U_4 & & & \\ & U_5 & & \\ & & U_6 & \\ & & & U_7 \end{bmatrix}}_U \underbrace{\left[\begin{array}{c|c} \tilde{A}_{45} & U_2 \tilde{A}_{23} V_3^* \\ \hline \tilde{A}_{54} & \tilde{A}_{67} \\ \hline U_3 \tilde{A}_{32} V_2^* & \tilde{A}_{76} \end{array} \right]}_{A^{(L-1)} \text{ HSS of size } 2^L r \ll n} \underbrace{\begin{bmatrix} V_4 & & & \\ & V_5 & & \\ & & V_6 & \\ & & & V_7 \end{bmatrix}}_V^*.$$



Standard telescopic decomposition

2 HSS matrices and telescopic decomposition

$$A = \underbrace{\begin{bmatrix} A_{44} & & & \\ & A_{55} & & \\ & & A_{66} & \\ & & & A_{77} \end{bmatrix}}_D + \underbrace{\begin{bmatrix} U_4 & & & \\ & U_5 & & \\ & & U_6 & \\ & & & U_7 \end{bmatrix}}_U \underbrace{\left[\begin{array}{c|c} \tilde{A}_{45} & U_2 \tilde{A}_{23} V_3^* \\ \hline \tilde{A}_{54} & \tilde{A}_{67} \\ \hline U_3 \tilde{A}_{32} V_2^* & \tilde{A}_{76} \end{array} \right]}_{A^{(L-1)} \text{ HSS of size } 2^L r \ll n} \underbrace{\begin{bmatrix} V_4 & & & \\ & V_5 & & \\ & & V_6 & \\ & & & V_7 \end{bmatrix}^*}_V.$$





Functions of a low-rank update

3 Low-rank updates and functions of HSS matrices

To approximate $f(B + \mathcal{E}) - f(B)$, Beckermann et al.⁶ take the following result

$$f\begin{pmatrix} B & \mathcal{E} \\ 0 & B + \mathcal{E} \end{pmatrix} = \begin{pmatrix} f(B) & f(B + \mathcal{E}) - f(B) \\ 0 & f(B + \mathcal{E}) \end{pmatrix}.$$

and project it onto small subspaces $\mathcal{W} \oplus \mathcal{Z} \subseteq \mathbb{C}^{2n}$ with bases W, Z :

$$f(B + \mathcal{E}) - f(B) \approx W \left[f \begin{pmatrix} W^* B W & W^* \mathcal{E} Z \\ 0 & Z^* B Z + Z^* \mathcal{E} Z \end{pmatrix} \right]_{12} Z^*.$$

⁶Beckermann, Kressner, and Schweitzer, *Low-rank updates of matrix functions*; Beckermann, Cortinovis, et al., *Low-rank updates of matrix functions II*.



Application to the leaf level

3 Low-rank updates and functions of HSS matrices

$$f(A) = f(D + UA^{(L-1)}V^*)$$

$$f(A) \approx f(D) + W_U \begin{bmatrix} I & 0 \end{bmatrix} f \left(\begin{bmatrix} W_U^* D W_U & W_U^* U A^{(L-1)} V^* W_V \\ 0 & W_V^* A W_V \end{bmatrix} \right) \begin{bmatrix} 0 \\ I \end{bmatrix} W_V^*,$$

where W_U, W_V are bases of block rational Krylov spaces $\mathcal{Q}^\square(D, U, \xi)$, $\mathcal{Q}^\square(D^*, V, \bar{\xi})$.



Application to the leaf level

3 Low-rank updates and functions of HSS matrices

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where W_U, W_V are bases of block rational Krylov spaces $\mathcal{Q}^\square(D, U, \xi)$, $\mathcal{Q}^\square(D^*, V, \bar{\xi})$.
Matrices are block-diagonal, and we are computing *small spaces*

$$\mathcal{Q}^\square(D, U, \xi) = \bigoplus_{\tau \in \mathcal{L}(\mathcal{T})} \mathcal{Q}^\square(D_\tau, U_\tau, \xi).$$



Recovering the telescopic structure

3 Low-rank updates and functions of HSS matrices

We can rewrite the argument of f as

$$\begin{bmatrix} W_U^* & \\ & W_V^* \end{bmatrix} \left(\begin{bmatrix} D & \\ & D \end{bmatrix} + \begin{bmatrix} U \\ U \end{bmatrix} A^{(L-1)} \begin{bmatrix} V \\ V \end{bmatrix}^* \right) \begin{bmatrix} W_U & \\ & W_V \end{bmatrix}.$$

Lemma

This matrix admits a telescopic decomposition if:

- the inner factor is in telescopic decomposition with respect to a tree $\mathcal{T}'$;
- the outer factors are block-diagonal and compatible with $\mathcal{T}'$.

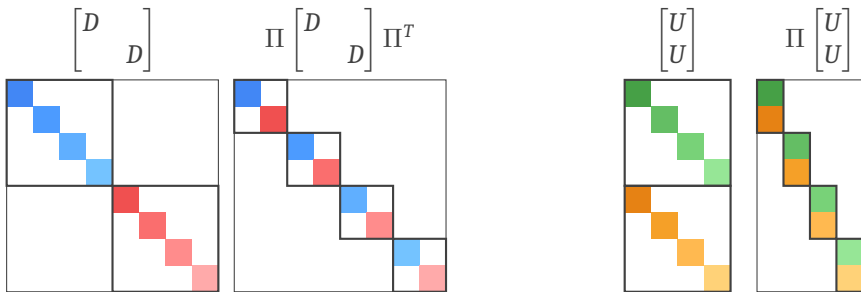
This is possible *up to a permutation* of rows and columns.



Block interlacing

3 Low-rank updates and functions of HSS matrices

$$A_{\text{aug}} = \tilde{\Pi} \begin{bmatrix} W_U^* & \\ & W_V^* \end{bmatrix} \Pi^T \left(\Pi \begin{bmatrix} D & \\ & D \end{bmatrix} \Pi^T + \Pi \begin{bmatrix} U \\ U \end{bmatrix} A^{(L-1)} \begin{bmatrix} V \\ V \end{bmatrix}^* \Pi^T \right) \Pi \begin{bmatrix} W_U & \\ & W_V \end{bmatrix} \tilde{\Pi}^T.$$





Telescopic decomposition of the augmented matrix

3 Low-rank updates and functions of HSS matrices

$$A_{\text{aug}} = D_{\text{aug}} + U_{\text{aug}} A^{(L-2)} V_{\text{aug}}^*.$$

Proposition

- has depth $L - 1$ if A has depth L ;
- the outermost level is $A^{(L-1)}$ modified by W_U and W_V ;
- the inner levels remain unchanged (telescopic decomposition of $A^{(L-2)}$).



Algorithm outline

3 Low-rank updates and functions of HSS matrices

1. **INPUT:** A Telescopic, f function, ξ_k complex poles;
2. **Bases:** compute W_U, W_V (RatKrylov on each diagonal block (D_i, U_i) (D_i^*, V_i));
3. **Diagonal part:** compute $f(D)$ directly ($f(D_i)$);
4. **Augmented matrix:** build the telescopic decomposition of A_{aug} ;
5. **Descend one level:** iterate on A_{aug} up to the root;
6. **Permutations:** pick submatrices of the outer level with $[I \ 0] \tilde{\Pi}^T$ and $\tilde{\Pi} [0 \ I]^T$;
7. **OUTPUT:** $f(A)$ Telescopic.



Cutting the outer level

3 Low-rank updates and functions of HSS matrices

$$f(A) \approx f(D) + [I \ 0] \tilde{\Pi}^T \underbrace{\left(f(D_{\text{aug}}) + W_{U_{\text{aug}}} [f(\cdots)]_{1,2} W_{V_{\text{aug}}}^* \right)}_{\approx f(A_{\text{aug}})} \tilde{\Pi} [0 \ I]^T.$$



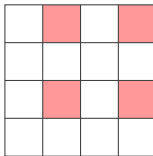
Cutting the outer level

3 Low-rank updates and functions of HSS matrices

$$f(A) \approx f(D) + [I \ 0] \tilde{\Pi}^T \underbrace{\left(f(D_{\text{aug}}) + W_{U_{\text{aug}}} [f(\cdots)]_{1,2} W_{V_{\text{aug}}}^* \right)}_{\approx f(A_{\text{aug}})} \tilde{\Pi} [0 \ I]^T.$$

The matrices $[I \ 0] \tilde{\Pi}^T$ and $\tilde{\Pi} [0 \ I]^T$ are block-diagonal with respect to $\mathcal{T}'$ and select the appropriate rows and columns to restore the original dimensions.

$$[I \ 0] \tilde{\Pi}^T f(D_{\text{aug}}) \tilde{\Pi} [0 \ I]^T \rightsquigarrow \begin{bmatrix} I & 0 \\ & I & 0 \end{bmatrix} [f(D_{\text{aug}})]_{i,i} \begin{bmatrix} 0 \\ I \\ & 0 \\ I \end{bmatrix}$$





Error estimate

4 Error Analysis and merging

Theorem (Global error bound)

Given A in telescopic decomposition with L levels, f “defined on the right matrices” and a number of poles m , the error of the algorithm satisfies

$$\|E_m^{(L)}(f)\|_F \leq (1 + \sqrt{2})^2 \min_{r \in \Pi_m/q_m} \|(f - r)'\|_{\infty, K^{(L)}} \sum_{i=1}^{L-1} \|A^{(i)}\|_F,$$

where $K^{(L)}$ is a compact convex set containing $W(A) + B\left(0, \sum_{i=1}^{L-1} 2^{\frac{i-3}{2}} \|A^{(i)}\|_2\right)$.



Merged basis

4 Error Analysis and merging

Projecting on a common basis $\text{span}(Q) = \text{span}[W_U, W_V]$ in the low-rank update allows to block-diagonalize A_{aug}

$$P^{-1}A_{\text{aug}}P = \begin{bmatrix} Q^*DQ & \\ & Q^*AQ \end{bmatrix}, \quad P := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \otimes I.$$

The approximation collapses to

$$f(M + E) - f(M) \approx Q \left[f(Q^*(M + E)Q) - f(Q^*MQ) \right] Q^*,$$

and maintains rational exactness. It improves the error estimate to

$$\mathcal{C} \min_{r \in \Pi_m/q_m} \|(f - r)'\|_{\infty, K^{(L)}} \sum_{i=1}^{L-1} \|A^{(i)}\|_F, \quad \text{where } K^{(L)} \supseteq W(A).$$



Exponential of random HSS matrices⁷ $\mathcal{O}(nr^2k^2)$

5 Experiments

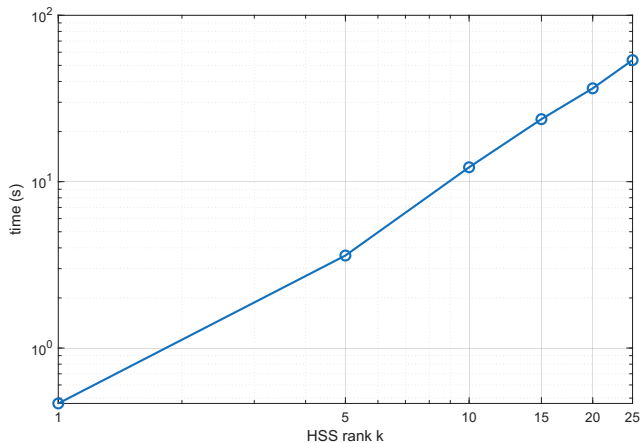
n	rel. err (Fro)	abs. err (Fro)	t_{Tel} (s)	t_{expm} (s)
512	2.143×10^{-16}	7.368×10^{-15}	0.18	0.16
1024	1.658×10^{-16}	6.518×10^{-15}	0.41	0.16
2048	1.651×10^{-16}	8.269×10^{-15}	0.79	0.73
4096	1.492×10^{-16}	1.003×10^{-14}	1.44	4.34
8192	2.288×10^{-14}	2.122×10^{-12}	1.92	26.24
16384	1.599×10^{-10}	2.072×10^{-08}	4.74	190.35

⁷Massei, Robol, and Kressner, *hm-toolbox: Matlab software for HODLR and HSS matrices*.



Quadratic in the HSS rank [1, 5, 10, 15, 20, 25]

5 Experiments





Advection-diffusion operator

5 Experiments

We compute $f(A) = A^{-1/2}$ for the one-dimensional A-D operator

$$\mathcal{L}u = -u'' + \eta u', \quad x \in (0, 1), \quad u(0) = u(1) = 0,$$

which yields the matrix

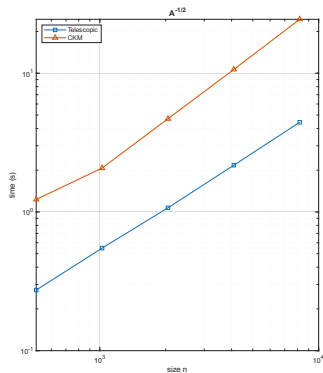
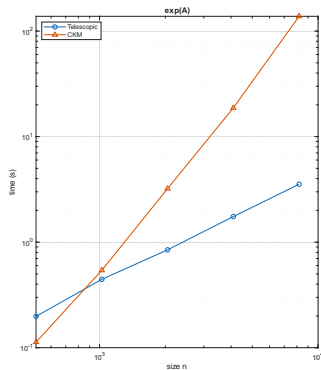
$$A_\eta = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix} + \frac{\eta}{2h} \begin{bmatrix} 0 & 1 & & \\ -1 & 0 & 1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 0 \end{bmatrix}.$$

Three regimes of increasing convection, $\eta = 10, 100, 1000$ and $n = 512, \dots, 4096$.



Telescopic vs CKM⁸ (1d A-D, $\eta = 100$)

5 Experiments

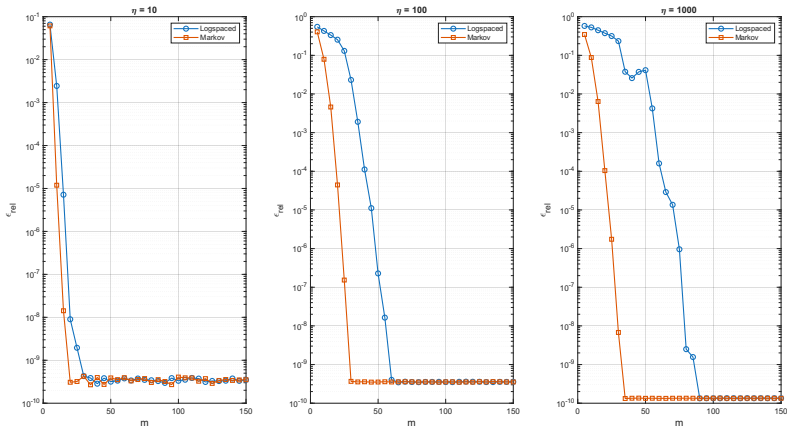


⁸Cortinovis, Kressner, and Massei, “Divide-and-conquer methods for functions of matrices with banded or hierarchical low-rank structure”.



Poles: logspaced vs Markov⁹

5 Experiments



⁹Beckermann and Reichel, “Error Estimates and Evaluation of Matrix Functions via the Faber Transform”.



Exponential integrators

5 Experiments

$$\begin{cases} u_t(x, t) = u_{xx}(x, t) + 10 u_x(x, t), & x \in (0, 1), t > 0, \\ u(0, t) = 0, \quad u(1, t) = 0, & t > 0, \\ u(x, 0) = u_0(x), & x \in [0, 1]. \end{cases}$$

Discretizing the spatial domain we obtain the ODE system $U'(t) = A U(t)$ where

$$A = (N + 1)^2 \text{ tridiag}(-1, 2, -1) + 5(N + 1) \text{ tridiag}(-1, 0, 1).$$



Exponential integrators

5 Experiments

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Two strategies:

- $t_n = n\Delta t$, $U^{n+1} = e^{\Delta t A} U^n \approx \text{expmv}(\Delta t, A, U^n)$;
- $U^{n+1} \approx \text{exp}_{\text{Tel}}(\tilde{A}) U^n$ where $\tilde{A} \approx A$ is an HSS approximation.

Total cost: $\mathcal{O}(nk^2) + T_c$ (one-time only compression).



expmv vs expTel

5 Experiments

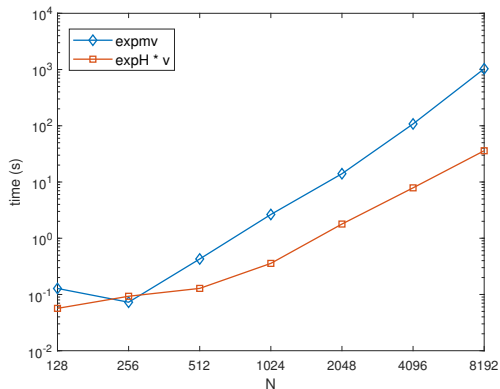


Figure: Time for 100 mat-vecs $\exp(tA)v$ with $t = 1e - 2$. We also include T_c .



Future work (a lot of it)

6 Conclusions

- Develop an adaptive error estimation strategy to select the number of poles m ;
- Propose more pole selection strategies (e.g., AAA on $W(T)$, T Toeplitz);
- Error estimates for Markov functions based on integrals of rational block FOM approximations;
- Extend this construction to other low (block) displacement rank matrices. (e.g., block Toeplitz).



Computing Functions of Rank-structured or Telescopic Matrices

Thank you for your attention!
Any questions?